

Amazon E-Commerce Database

Chapter – 2

2.0 Introduction

The entities required for the Amazon E-Commerce Database are identified based on the business requirements of an online shopping platform. The attributes of each entity, along with their Primary Keys, Foreign Keys, Unique Keys, and constraints, are identified to organize data efficiently. The relationships between the entities represent customer purchases, product management, order processing, payment handling, shipping, and product reviews. This analysis forms the basis for designing the ER Diagram, Relational Schema, and Database Implementation while maintaining data integrity, consistency, and minimal redundancy.

1. Customer

Attribute	Key Type
Customer ID	Primary Key
First Name	Not Null
Last Name	Not Null
Email	Unique Key
Phone Number	Unique
Password	Not Null
Gender	-
Date of Birth	-
Address	-
City	-
State	-
Country	-
Pincode	-
Registration Date	-

2. Seller

Attribute	Key Type
Seller ID	Primary Key
Seller Name	Not Null
Business Name	Not Null
Email	Unique Key
Phone Number	Unique
GST Number	Unique
Address	-
Rating	-

3. Product

Attribute	Key Type
Product ID	Primary Key
Product Name	Not Null
Seller ID	Foreign Key
Category ID	Foreign Key
Brand ID	Foreign Key
Price	Not Null
Discount	-
Stock	Not Null
Weight	-
Color	-
Warranty	-
Description	-
Rating	-

4. Category

Attribute	Key Type
Category ID	Primary Key
Category Name	Unique Key
Description	-

5. Brand

Attribute	Key Type
Brand ID	Primary Key
Brand Name	Unique Key
Country	-

6. Cart

Attribute	Key Type
Cart ID	Primary Key
Customer ID	Foreign Key
Total Items	-
Grand Total	-

7. Cart_Item

Attribute	Key Type
CartItem ID	Primary Key
Cart ID	Foreign Key
Product ID	Foreign Key
Quantity	-

Price	-
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8. Order

Attribute	Key Type
Order ID	Primary Key
Customer ID	Foreign Key
Order Date	-
Delivery Date	-
Order Status	-
Total Amount	-

9. Order_Item

Attribute	Key Type
OrderItem ID	Primary Key
Order ID	Foreign Key
Product ID	Foreign Key
Quantity	-
Unit Price	-

10. Payment

Attribute	Key Type
Payment ID	Primary Key
Order ID	Foreign Key
Payment Method	-
Payment Status	-
Payment Date	-
Transaction ID	Unique Key
Amount	-

11. Shipment

Attribute	Key Type
Shipment ID	Primary Key
Order ID	Foreign Key
Delivery Partner ID	Foreign Key
Tracking Number	Unique Key
Shipment Date	-
Delivery Date	-
Shipment Status	-

12. Delivery Partner

Attribute	Key Type
Delivery Partner ID	Primary Key
Partner Name	Not Null
Contact Number	-
Vehicle Number	-

13. Warehouse

Attribute	Key Type
Warehouse ID	Primary Key
Warehouse Name	-
Location	-
Capacity	-

14. Inventory

Attribute	Key Type
Inventory ID	Primary Key
Product ID	Foreign Key
Warehouse ID	Foreign Key
Available Stock	-
Last Updated	-

15. Review

Attribute	Key Type
Review ID	Primary Key
Customer ID	Foreign Key
Product ID	Foreign Key
Rating	Check (1–5)
Review Text	-
Review Date	-

16. Wishlist

Attribute	Key Type
Wishlist ID	Primary Key
Customer ID	Foreign Key
Product ID	Foreign Key
Added Date	-

17. Return

Attribute	Key Type
Return_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Return_Date	-
Return_Reason	-
Refund_Status	-

18. Coupon

Attribute	Key Type
Coupon_ID	Primary Key
Coupon_Code	Unique Key
Discount_Percentage	-
Expiry_Date	-
Minimum_Order_Value	-

2.2 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial / Total
Customer	Places	Order	One-to-Many	Partial / Total
Customer	Writes	Review	One-to-Many	Partial / Total
Seller	Sells	Product	One-to-Many	Total / Total
Category	Categorizes	Product	One-to-Many	Total / Total
Cart	Contains	Cart Item	One-to-Many	Total / Total
Cart Item	Refers To	Product	Many-to-One	Total / Partial
Order	Includes	Order Item	One-to-Many	Total / Total
Order Item	Refers To	Product	Many-to-One	Total / Partial
Order	Has	Payment	One-to-One	Total / Total
Order	Has	Shipment	One-to-One	Total / Total
Product	Receives	Review	One-to-Many	Partial / Total